

Disproof of conjecture 00000001729:
the Campana count of $(\mathbb{P}^1, \frac{1}{2}[0] + \frac{1}{2}[1] + \frac{1}{2}[\infty])$ is not $c\sqrt{B}/\sqrt{\log B}$

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1 The conjecture and the reading

Statement (English, verbatim). Definition: Campana points are rational points satisfying orbifold denominator divisibility conditions. Conjecture: The count of Campana points of $(\mathbb{P}^1, 2[0] + 2[1] + 2[\infty])$ is always c times the square root of B divided by the square root of $\log B$; and the logarithmic correction is given by the additive-energy duality. (The Chinese text states the same: the count is always $c \cdot B^{1/2}/(\log B)^{1/2}$, and the logarithmic correction is given by additive-energy duality.)

Reading. The notation $2[0] + 2[1] + 2[\infty]$ assigns multiplicity $m = 2$ to each of the points $0, 1, \infty$; as a Campana orbifold divisor this is $\Delta = \frac{1}{2}[0] + \frac{1}{2}[1] + \frac{1}{2}[\infty]$ (a coefficient 2 itself is not an orbifold coefficient). The conjecture is a conjunction. We refute its first conjunct,

$$(A) \quad N(B) \text{ “is” } c\sqrt{B}/\sqrt{\log B},$$

which refutes the conjunction. The second conjunct (“additive-energy duality”) is undefined in the text; it is not used and not addressed. We refute (A) under each of the following readings, for every real constant c :

- (exact) $N(B) = c\sqrt{B}/\sqrt{\log B}$ for all $B \geq B_0$ (any B_0 ; “always”);
- (asymptotic) $N(B) \sim c\sqrt{B}/\sqrt{\log B}$ as $B \rightarrow \infty$;
- (order of magnitude) $N(B) = O(\sqrt{B}/\sqrt{\log B})$; this also covers $N(B) \asymp \sqrt{B}/\sqrt{\log B}$.

We do this for the naive height H and for every height H^κ with $0 < \kappa \leq 1$, which includes the orbifold anticanonical height $H^{1/2}$ (on $\mathbb{P}^1$, $-(K + \Delta)$ has degree $2 - \frac{3}{2} = \frac{1}{2}$). *Not refuted:* the one-sided reading $N(B) \gg \sqrt{B}/\sqrt{\log B}$ (our lower bound implies it), and heights H^κ with $\kappa > 1$.

2 Definitions (as formalized)

Conventions. A point of $\mathbb{P}^1(\mathbb{Q})$ outside the support $\{0, 1, \infty\}$ of Δ is a rational number $x \neq 0, 1$. We write $x = a/b$ in lowest terms with $b > 0$ (Lean `x.num`, `x.den`). On the standard model $\mathbb{P}_{\mathbb{Z}}^1$ the p -adic intersection numbers of x with the closures of $[0]$, $[\infty]$, $[1]$ are $v_p(a)$, $v_p(b)$ and $v_p(a - b)$. Valuations of integers are Mathlib’s `padicValInt` (of $|n|$).

Definition 1 (Campana condition, multiplicity 2). An integer n satisfies `Mult2` if for every prime p , $v_p(n) = 0$ or $v_p(n) \geq 2$.

Definition 2 (Campana point). $x \in \mathbb{Q}$ is a Campana point of $(\mathbb{P}^1, \frac{1}{2}[0] + \frac{1}{2}[1] + \frac{1}{2}[\infty])$ (`IsCampanaPoint`) if $x \neq 0$, $x \neq 1$, and $a, b, a - b$ all satisfy `Mult2`. That is: for each of the three divisors and each prime p , the intersection number is 0 or at least the multiplicity 2. Equivalently, $|a|, b, |a - b|$ are squareful (powerful) and $\gcd(a, b) = 1$.

Definition 3 (Height and count). The naive height is $H(a/b) = \max(|a|, b)$ (`height`). For real B , $N(B) = \#\{x : x \text{ Campana}, H(x) \leq B\}$ (`campanaCount`, a `Set.ncard`; the set is finite by `finite_height_le`), and $N_\kappa(B) = \#\{x : x \text{ Campana}, H(x)^\kappa \leq B\}$ (`campanaCountPow`; finite for $\kappa > 0$ by `finite_pow`).

3 The proof

Lemma 4 (`pt_num_den, pt_campana, pt_inj`). Let m, n be positive integers with $\gcd(m, n) = 1$, m odd and n even. Put $x = (m^2 + n^2)^2 / (m^2 - n^2)^2$. Then x is in lowest terms as written, i.e. $a = (m^2 + n^2)^2$, $b = (m^2 - n^2)^2$, and $a - b = (2mn)^2$; x is a Campana point; and (m, n) is determined by x .

Proof. $m^2 - n^2$ is odd, hence nonzero. By the classification of primitive Pythagorean triples (`Mathlib.PythagoreanTriple.coprime_classification`, used in the easy direction), $\gcd(m^2 - n^2, 2mn) = 1$. Hence $(m^2 - n^2)^2$ is coprime to $(2mn)^2 + (m^2 - n^2)^2 = (m^2 + n^2)^2$, so the fraction is reduced. Each of $a, b, a - b$ is a nonzero square k^2 , and $v_p(k^2) = 2v_p(k) \in \{0\} \cup [2, \infty)$. $x \neq 0$ as $a \neq 0$; $x \neq 1$ since otherwise $a = b = 1$ and $(2mn)^2 = a - b = 0$. If two such pairs give the same x , then $m^2 + n^2 = m'^2 + n'^2$ and $m^2 - n^2 = \pm(m'^2 - n'^2)$. The sign $+$ gives $m = m', n = n'$; the sign $-$ gives $m = n'$, impossible by parity. $\square$

Lemma 5 (`card_good`). For every $K \geq 0$, the set G_K of pairs (i, j) with $0 \leq i < K$, $1 \leq j \leq K$ and $\gcd(2i + 1, 2j) = 1$ has at least $K^2/2$ elements.

Proof. There are K^2 pairs. If $\gcd(2i + 1, 2j) = g \neq 1$, then g is odd (it divides $2i + 1$) and $3 \leq g \leq 2K$, so $g = 2e + 3$ with $0 \leq e < K$, and $d = 2e + 3$ divides both $2i + 1$ and $2j$. For fixed odd d , the map $i \mapsto 2i + 1$ injects $\{i < K : d \mid 2i + 1\}$ into the multiples of d in $(0, 2K]$, of which there are $\lfloor 2K/d \rfloor$ (`Mathlib.Nat.Ioc_filter_dvd_card_eq_div`); and $d \mid 2j$ iff $d \mid j$, giving at most $\lfloor K/d \rfloor$ values of j . So at most $2K^2/d^2$ bad pairs are divisible by d , and the number of bad pairs is at most

$$2K^2 \sum_{e=0}^{K-1} \frac{1}{(2e+3)^2} \leq 2K^2 \left(\frac{1}{4} - \frac{1}{4(K+1)} \right) \leq \frac{K^2}{2},$$

by the telescoping bound $\frac{1}{(2e+3)^2} \leq \frac{1}{4(e+1)} - \frac{1}{4(e+2)}$ (`sum_inv_sq_odd`, proved by induction), which holds because $(2e + 3)^2 \geq 4(e + 1)(e + 2)$. $\square$

Theorem 6 (`campanaCount_lower`). For every $K \in \mathbb{N}$, $N(64K^4) \geq K^2/2$; that is, $N(B) \geq \sqrt{B}/16$ along $B = 64K^4$.

Proof. Map $(i, j) \in G_K$ to the point x of Lemma 1 with $m = 2i + 1$, $n = 2j$. It is injective and lands in Campana points with $H(x) = (m^2 + n^2)^2 \leq (8K^2)^2 = 64K^4$, since $m^2 + n^2 \leq (2K)^2 + (2K)^2$. Apply Lemma 2. $\square$

Theorem 7 (`campana_count_disproof, campana_count_pow_disproof`). For the naive height, and for N_κ with $0 < \kappa \leq 1$: $N(B)$ is not $O(\sqrt{B}/\sqrt{\log B})$ as $B \rightarrow \infty$ over the reals; for no real c is $N(B) \sim c\sqrt{B}/\sqrt{\log B}$; and for no real c, B_0 is $N(B) = c\sqrt{B}/\sqrt{\log B}$ for all $B \geq B_0$.

Proof. (`not_isBig0_of_lower`) Suppose $|N(B)| \leq C \sqrt{B}/\sqrt{\log B}$ for $B \geq B_0$. Choose an integer $K \geq 1$ with $K > \max(e^{256C^2}, B_0)$ and put $B = 64K^4 \geq B_0$. Then $\sqrt{B} = 8K^2$ and $s = \sqrt{\log B} \geq \sqrt{\log K} > 16|C|$, so $K^2/2 \leq N(B) \leq 8CK^2/s < K^2/2$, a contradiction. (`readings_of_not_isBig0`) An eventual equality $N = cg$ or an equivalence $N \sim cg$ implies $N = O(cg) = O(g)$ with $g = \sqrt{B}/\sqrt{\log B}$. For N_κ : if $H(x) \leq B$ and $B \geq 1$ then $H(x)^\kappa \leq B^\kappa \leq B$, so $N(B) \leq N_\kappa(B)$ (`count_le_countPow`), and the lower bound transfers. $\square$

4 Lean formalization and axiom audit

File `lean/Conjecture1729/Basic.lean` (370 lines, only import `Mathlib`; identical to the checked file), namespace `C1729`. Main statements:

```
theorem campana_count_disproof :
  (not exists c B0 : Real, forall B >= B0,
    (campanaCount B : Real) = c * (sqrt B / sqrt (log B))) /\
  (forall c : Real, not (fun B => (campanaCount B : Real)) ~[atTop]
    (fun B => c * (sqrt B / sqrt (log B)))) /\
  not (fun B => (campanaCount B : Real)) =0[atTop]
  (fun B => sqrt B / sqrt (log B))
theorem campana_count_pow_disproof (k : Real) (hk : 0 < k) (hk1 : k <= 1) :
  (the same three statements for campanaCountPow k)
theorem campanaCount_lower (K : Nat) :
  (K : Real) ^ 2 / 2 <= campanaCount (64 * (K : Real) ^ 4)
```

(ASCII rendering: in the Lean source, `not`, `exists`, `forall`, `/\` are the logical symbols, `Real` and `Nat` are $\mathbb{R}$ and $\mathbb{N}$, `sqrt` is `Real.sqrt`, `log` is `Real.log`, and the exponent is named κ .) `lake build` succeeds. `#print axioms` for each of the three theorems reports [`propext`, `Classical.choice`, `Quot.sound`]. There is no `sorry`, no `native_decide` and no added axiom.

5 Remarks

- The points used have $a > b > 0$, so they are also counted by any variant restricted to positive squareful triples $b + (a - b) = a$ or to $x > 1$. All of $a, b, a - b$ are perfect squares, so the same points are also Darmon points (valuations divisible by 2); this variant is not formalized.
- Context. Browning and Van Valckenborgh, *Sums of three squareful numbers* (<https://arxiv.org/abs/1106.4472>), “investigate the frequency of positive squareful numbers x, y, z for which $x+y=z$ ” and “examine the quantitative arithmetic of integral points on certain Campana orbifolds”; they note that a lower bound follows “via the familiar parametrisation for Pythagorean triples”. The argument above is an elementary, self-contained version of that observation; nothing from that paper is used in the proof.